THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS V — MATHEMATICS · MODEL PAPER 02 SOLVED

Learning-Level Evaluation Pattern (RSCERT) | Session 2026–27

Time: 2.30 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

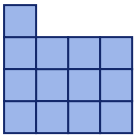
General Instructions:

1. This question paper is based on the learning areas of Class 5 Mathematics (Shape and Space, Number System, Operations, Measurement, Data Handling and Patterns).
2. Write your Name and Roll No. clearly on the question paper / answer sheet.
3. All questions are compulsory.
4. The marks allotted to each question are given against it.
5. Write your answers neatly and show the steps of working wherever required.

UNDERSTANDING SHAPE AND SPACE — Marks 21

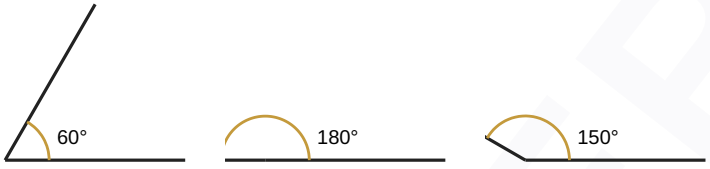
1. Each small square is 1 cm × 1 cm. What is the area of the shaded shape? [1]

(a) 12 sq cm **(b)** 16 sq cm **(c)** 26 sq cm **(d)** 13 sq cm



Ans. (d) 13 sq cm

2. Three angles P, Q and R are shown with their measures. [6]



(i) Write the type of angle: (a) P _____ (b) Q _____ (c) R _____ 3

Ans. (a) acute (b) straight (c) obtuse

(ii) Which solid is formed by folding each net? (a) a net of 6 equal squares _____ (b) a net of 2 circles and 1 rectangle _____ (c) a net of 1 circle and 1 sector of a circle _____ 3

Ans. (a) cube (b) cylinder (c) cone

3. [6]

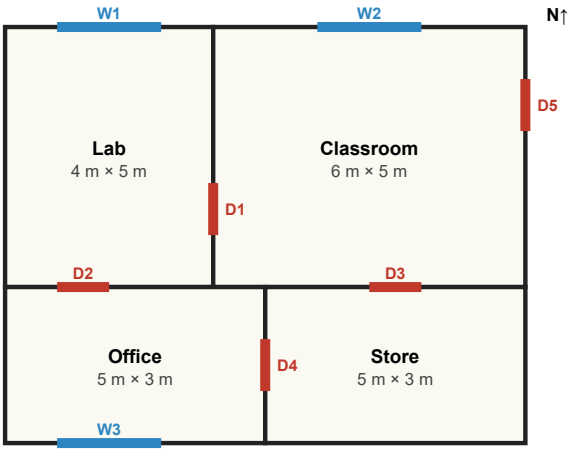
(i) (a) How many vertices does a cuboid have? (b) How many edges does a triangular pyramid (tetrahedron) have? (c) How many flat faces does a cylinder have? 3

Ans. (a) 8 (b) 6 (c) 2

(ii) Name the solid that each object looks like: (a) a matchbox (b) a football (c) a joker's cap 3

Ans. (a) cuboid (b) sphere (c) cone

4. The plan of a school block in Alwar is shown. Study it and answer. [8]



D = door (red), W = window (blue).

(a) What is the area of the Classroom? 2

Ans. 6 × 5 = **30 sq m**

(b) How much more floor area does the Classroom have than the Lab? 2

Ans. 30 − 20 = **10 sq m**

(c) The Office floor is to be covered with carpet at ₹ 40 per sq m. Find the cost. 2

Ans. 15 × 40 = **₹ 600**

(d) Find the perimeter of the whole block (10 m by 8 m). How many doors are there in all? 2

Ans. 2 × (10 + 8) = **36 m**; doors = **5**

NUMBER SYSTEM — Marks 9

5. The number 3,45,200 written in the international system is [1]

(a) 3,45,200 **(b)** 345,200 **(c)** 34,520 **(d)** 3,452,000

Ans. (b) 345,200

6. The successor of 99,999 is [1]

(a) 99,990 **(b)** 10,000 **(c)** 1,00,000 **(d)** 10,00,000

Ans. (c) 1,00,000

7. Which is the smallest of these numbers: 48,210; 48,021; 48,201; 48,012? [1]

(a) 48,021 **(b)** 48,201 **(c)** 48,210 **(d)** 48,012

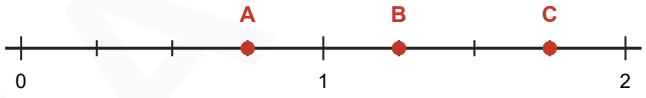
Ans. (d) 48,012

8. [6]

(i) Ravi walked 1/2 km in the morning and 1/4 km in the evening. (a) How far did he walk in all? (b) How much less than 1 km is that? 3

Ans. (a) 1/2 + 1/4 = 2/4 + 1/4 = **$\frac{3}{4}$ km** (b) 1 − 3/4 = **$\frac{1}{4}$ km**

(ii) Write the fraction shown by A, B and C: A = ____ , B = ____ , C = ____ 3



Ans. A = $\frac{3}{4}$, B = $1\frac{1}{4}$, C = $1\frac{3}{4}$

OPERATIONS — Marks 25

9. 36 × 25 = ? [1]

(a) 725 **(b)** 900 **(c)** 720 **(d)** 1000

Ans. (b) 900

10. A dairy has 40 litres of milk. It fills 8 bottles of 750 mL each. How many litres of milk are left? [3]

Ans. Milk in bottles = 8 × 750 = 6000 mL = 6 L
Left = 40 − 6 = **34 L**

11. A school library has 3,458 books. They are arranged equally on 8 shelves. How many books are on each shelf and how many are left over? Verify your answer. [3]

Ans. 3458 ÷ 8: Quotient = **432** books per shelf, Remainder = **2**
Check: 8 × 432 + 2 = 3456 + 2 = 3458 ✓

12. [6]

(i) Anita bought 5 kg rice at ₹ 62 per kg, 2 kg dal at ₹ 118 per kg and 1 L oil for ₹ 145. She paid ₹ 1,000. How much change did she get? 3

Ans. Rice = ₹ 310; Dal = ₹ 236; Oil = ₹ 145; Bill = ₹ 691
Change = 1000 − 691 = **₹ 309**

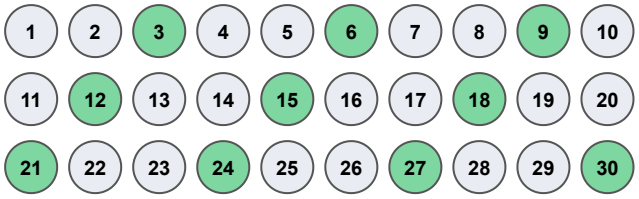
(ii) (a) Estimate 6,812 − 2,395 by rounding each number to the nearest hundred. (b) Find the exact difference. 3

Ans. (a) 6,800 − 2,400 = **4,400** (b) 6,812 − 2,395 = **4,417**

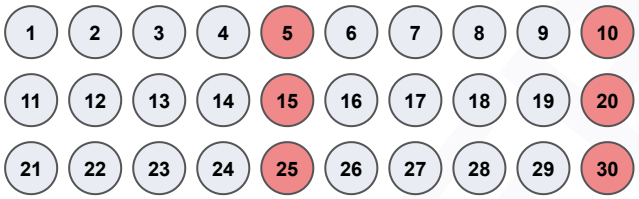
13. [6]

(i) Bulbs 1 to 30 are connected to two switches P and Q. The bulbs that glow are coloured. (a) How many bulbs glow with switch P? (b) How many glow with switch Q? (c) Which bulbs glow with both switches? 3

Switch P



Switch Q



Ans. (a) **10** (b) **6** (c) **15 and 30** (common multiples of 3 and 5)

(ii) Two school bells ring together at 8:00 a.m. One rings every 12 minutes and the other every 18 minutes. (a) After how many minutes will they ring together again? (b) At what time? 3

Ans. (a) LCM of 12 and 18 = **36 minutes** (b) 8:00 + 36 min = **8:36 a.m.**

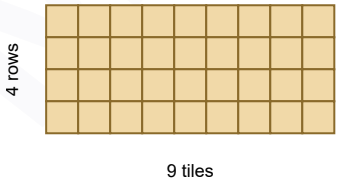
14. [6]

(i) In this filled cross-number grid, each ROW is an ACROSS number and each COLUMN is a DOWN number. (a) Write the smallest ACROSS number. (b) Add the 1st and 2nd ACROSS numbers. (c) Write the greatest DOWN number. 3

3	6	1
8	5	2
9	4	7

Ans. (a) **361** (b) 361 + 852 = **1213** (c) **654**

(ii) 36 square tiles of side 1 cm are arranged as a rectangle of 9 tiles in a row (figure). (a) How many rows are there? (b) Find the perimeter of the rectangle. (c) If the same tiles are put in 6 rows, how many tiles are in each row? 3



Ans. (a) 36 ÷ 9 = **4 rows** (b) 2 × (9 + 4) = **26 cm** (c) 36 ÷ 6 = **6 tiles**

MEASUREMENT — Marks 17

15. Fill in the blanks: (a) 2.5 km = _____ m (b) 4500 g = _____ kg (c) 2 h 15 min = _____ min [3]

Ans. (a) 2500 (b) 4.5 (c) 135

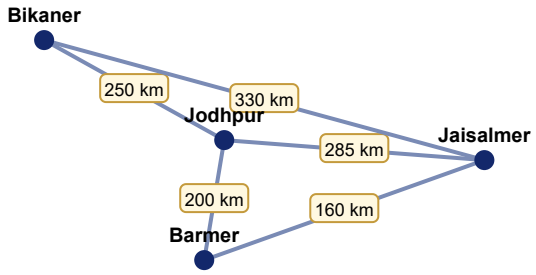
16. Write True or False: (a) 3 p.m. is 13:00 on a 24-hour clock. (b) The normal temperature of the human body is about 37 °C. (c) ₹ 3.05 = 305 paise. [3]

Ans. (a) False (3 p.m. = 15:00) (b) True (c) True

17. A trader bought 2 sacks of wheat of 48 kg 500 g each and 1 sack of bajra of 36 kg 750 g. Find the total mass of grain in kg and g. [3]

Ans. Wheat = 2 × 48 kg 500 g = 97 kg
Total = 97 kg + 36 kg 750 g = **133 kg 750 g**

18. Distances between four cities of Rajasthan are shown on the map (not to scale). [8]



(a) Find the distance from Jodhpur to Jaisalmer through Barmer. 2

Ans. 200 + 160 = **360 km**

(b) How much shorter is the direct Jodhpur–Jaisalmer road than the route through Barmer? 2

Ans. 360 − 285 = **75 km**

(c) Which city is nearest to Jodhpur? 2

Ans. Barmer (200 km)

(d) A car goes Bikaner → Jodhpur → Barmer → Jaisalmer. How far does it travel? How much more is it than the direct road Bikaner–Jaisalmer? 2

Ans. 250 + 200 + 160 = **610 km**; 610 − 330 = **280 km** more

DATA HANDLING AND PATTERNS — Marks 8

19. Which of these is a square number? [1]

(a) 120 **(b)** 122 **(c)** 121 **(d)** 125

Ans. (c) 121

20. Squares are made in a row with matchsticks. Fig. 1 needs 4 sticks. How many sticks are needed for Fig. 6? [1]

(a) 19 **(b)** 16 **(c)** 18 **(d)** 21



Fig. 1 Fig. 2 Fig. 3

Ans. (a) 19

21. Students of Class 5 were asked to choose their favourite fruit. The tally table shows the result (a group of five is shown as ||||). [3]

Fruit	Tally marks
Mango	𝍸 𝍸 |||
Apple	𝍸 ||||
Banana	𝍸 ||
Guava	𝍸 |
Orange	𝍸

(a) Which fruit was chosen by the most students? How many chose it? 1

Ans. Mango, 13 students

(b) How many students were asked in all? 1

Ans. 13 + 9 + 7 + 6 + 5 = **40**

(c) How many more students chose Mango than Banana? 1

Ans. 13 − 7 = **6**

22. Look at the pattern: 1 = 1 × 1; 1 + 3 = 2 × 2; 1 + 3 + 5 = 3 × 3. Find: (a) 1 + 3 + 5 + 7 (b) 1 + 3 + 5 + 7 + 9 + 11 (c) the sum of the first 10 odd numbers [3]

Ans. The sum of the first n odd numbers = n × n. (a) 4 × 4 = **16** (b) 6 × 6 = **36** (c) 10 × 10 = **100**

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND